



**Assignment 5: Analysis and Simulation of Single and Dual Active
Bridge Converters for Auxiliary Power Modules**

ELEC 8900-58: EV Power Conversion

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July 28th, 2025

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I. INTRODUCTION

In Electric vehicles there is a large amount of power electronic converters that can manage energy between their battery packs and their other subsystems. One of these critical subsystems is the Auxiliary Power Module (APM), responsible for stepping down the high DC voltage (400-800V) to 12 V to supply power to low voltage systems including: sensors, lighting, power outlets, infotainment systems, etc. To satisfy the requirements of efficiency, size and safety of passengers, often these modules exploit high frequency transformer-based DC-DC converters.

A very widely used topology in the industry that meets these requirements is the Dual Active bridge (DAB). It has the ability not only to transfer energy efficiently, but also to provide bidirectional energy flow between the battery packs and the APM, exploit soft switching, and provide full isolation. The DAB consists of two full active bridges inverters both separated by a high frequency transformer. The way power flow is controlled is by using phase shift modulation.

The main objective of this report is to explore the performance and behavior of the DAB converter through a preconfigured Simulink model provided in the course; The model consists of a 400:12 V high frequency transformer, inductive and capacitive impedance compensation, and a control loop using a PI controller to perform phase shift regulation. In addition to that, the converter is modified to become a Single Active Bridge (SAB) by removing the secondary bridge and replacing it with a diode rectifier. The analysis will compare SAB and DAB and will investigate soft switching.

II. DISCUSSION

A. Dual Active Bridge Converter

1) *Theory of Operation*

An isolated DC-DC converter known as a Dual Active Bridge (DAB) is mainly used when energy needs to be transferred between two distinct voltage levels, such as in electric vehicles when a 400 V battery needs to be connected to a 12 V system. A high-frequency transformer, an H-bridge on the low-voltage side, and an H-bridge on the high-voltage side comprise its three primary components.

Both H-bridges produce square wave voltages. The phase shift between these two waveforms is the only means by which to control the quantity of power transferred. When we provide a phase shift between input and output bridges, it creates a voltage difference across the transformer's inductance. The current flows due to it, and hence the power is transferred.

2) *Core Principle*

Both bridges are fed with high-frequency square waves combined. Isolation and (with its leakage inductance) restriction of current rise are made by the transformer. Phase shift between the bridges' square waves controls magnitude and direction of transferred power.

When the two sides' voltage square waves are exactly in phase, the average voltage across the transformer's leakage inductance is zero, and so no net power gets transferred. A phase shift introduces a nonzero average voltage across the inductor, causing current flow and therefore power transfer.

3) *Bidirectional Control*

If the leading bridge is on the primary side, power transfers from the primary to the secondary.

If the leading bridge is on the secondary side, the direction is reversed.

This property allows both power transfer for charging and vehicle-to-grid (V2G) application.

4) *PWM Phase Shift Modulation*

Your simulation's control block uses the current regulator to create a reference current.

PI control controls the required phase shift (δ), sent to the phase-shifted gating logic to modulate power transfer directly

To make this clear, the transformer and leakage inductance can be represented by a resistor and inductor in series RL circuit. Both H-bridges act as an AC voltage source. Thus, the circuit is two AC sources with a phase difference joined by an RL branch.

Below is a representative circuit of a DAB

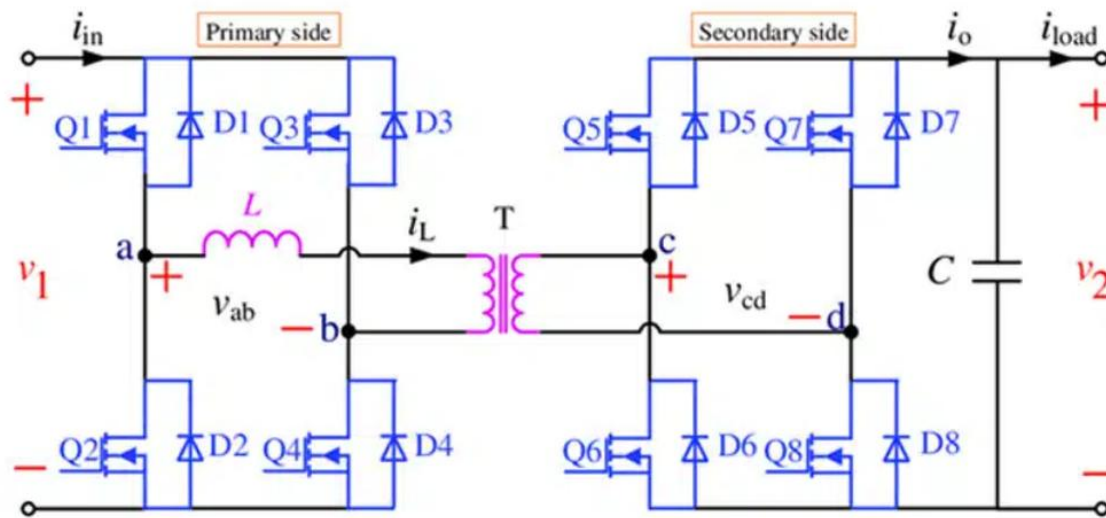


Figure 1: Dual Active bridge Block Diagram

5) Mathematical Model

From the lecture, and again supposing primarily inductive action (which is most typical), the present via the transformer simplifies to:

$$i = \frac{V_1}{\omega L} \left(\sin \delta + j \left(\frac{V_2}{V_1} - \cos \delta \right) \right)$$

This leads to the power formula:

$$P = \frac{V_1 V_2}{2\omega L} \sin \delta$$

This equation tells us that the larger we make the phase shift, the more power that we pass across. Power flows from the high voltage to the low voltage if the phase shift is positive. It flows the other way if it's negative. This enables DAB to pass power in both directions.

6) Soft Switching

What is Soft Switching?

New power conversion systems increasingly rely on soft switching to reduce energy loss, improve thermal efficiency, and support high-frequency operation. In traditional hard-switched systems, power semiconductors like MOSFETs or IGBTs are turned on and off when voltage and current are simultaneously nonzero. This causes high switching losses, electromagnetic interference (EMI), and increased thermal stress on devices [1].

Soft switching avoids these problems by allowing the switching to be performed in near-zero voltage or current conditions:

Zero Voltage Switching (ZVS) occurs at turning on the switch voltage is nearly zero.

Zero Current Switching (ZCS) occurs at turning on when the current through the switch is nearly zero.

These methods result in:

- Increase the lifetime value of the components
- Less power losses
- Less EMI
- Higher switching frequency leads to more compact design

The Operation of Soft Switching in DAB Converters

One of the Dual Active Bridge (DAB) converter's inherent characteristics of being able to provide soft switching under specific circumstances is its biggest strength. Current starts flowing through the transformer before the switches are triggered by the phase shift between the secondary and main bridges. Before the switching, the initial current flow brings the voltage across the switches down to zero, enabling the converter to enter Zero Voltage Switching (ZVS) mode. Lower voltage stress initiates the switching, lessening the generation of heat and switching loss [1]. This is intrinsic when the system has:

- An adequate load
- Moderate or high phase shift
- An inductive profile

Lower temperatures enable simpler and more efficient switching, reducing the need for complex heat management systems and enhancing reliability.

Advantages of Soft Switching

The use of soft switching technology in EVs has numerous advantages:

- Higher efficiency due to a dramatic reduction in switching loss, particularly at higher frequencies.
- Reduced heat generation enables thermal miniaturization and heat sinks, resulting in thermal reliability.
- As smooth switching eliminates high-frequency noise and the need for harsh filtering, electromagnetic interference (EMI) is reduced.
- Small switching frequencies necessitate minuscule capacitors and inductors, and this results in a compact structure.
- Complete bidirectionality in charging applications eliminates hardware reconfiguration needs by providing smooth transitions between charging and discharging modes.

Overall, soft switching is an essential aspect of EVs that supports high efficiency and two-way power delivery. Specifically, the SS-DAB topology makes it possible to achieve the promise of high efficiency and a compact circuit. With proper phase shift control and implementation of natural inductive component phenomena, near-zero voltage switching is achievable, thereby optimizing performance [1].

7) DAB implementation

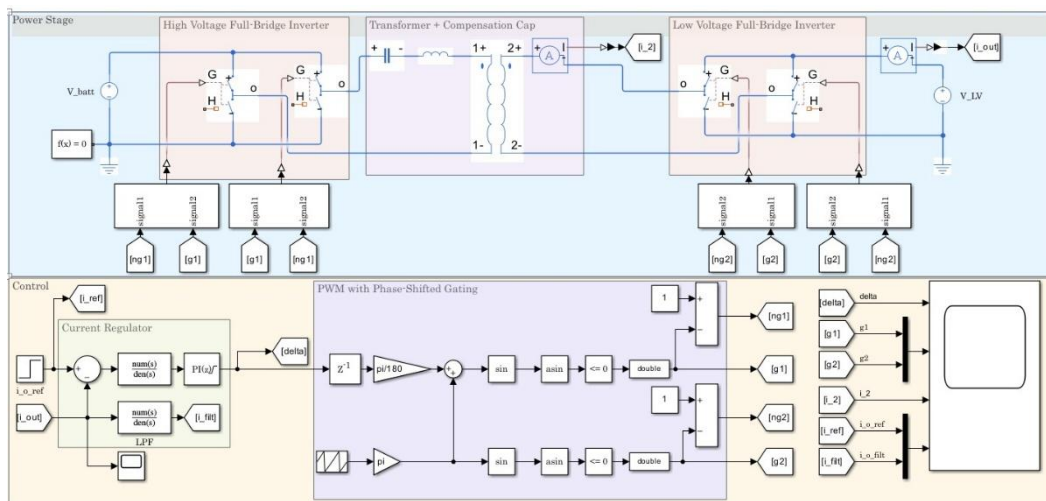


Figure 2: DAB implementation in Simulink

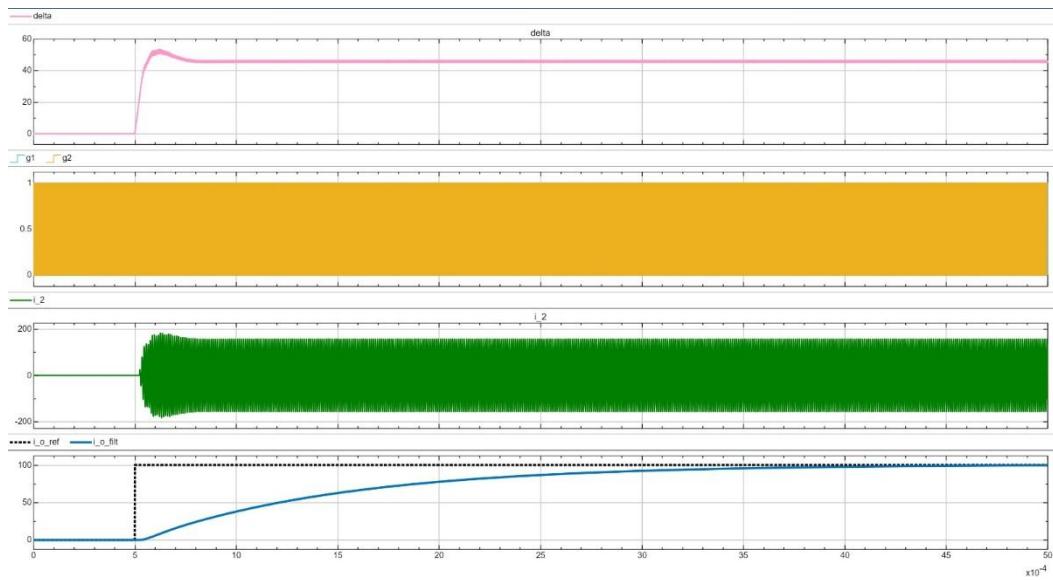


Figure 3: DAB waveforms

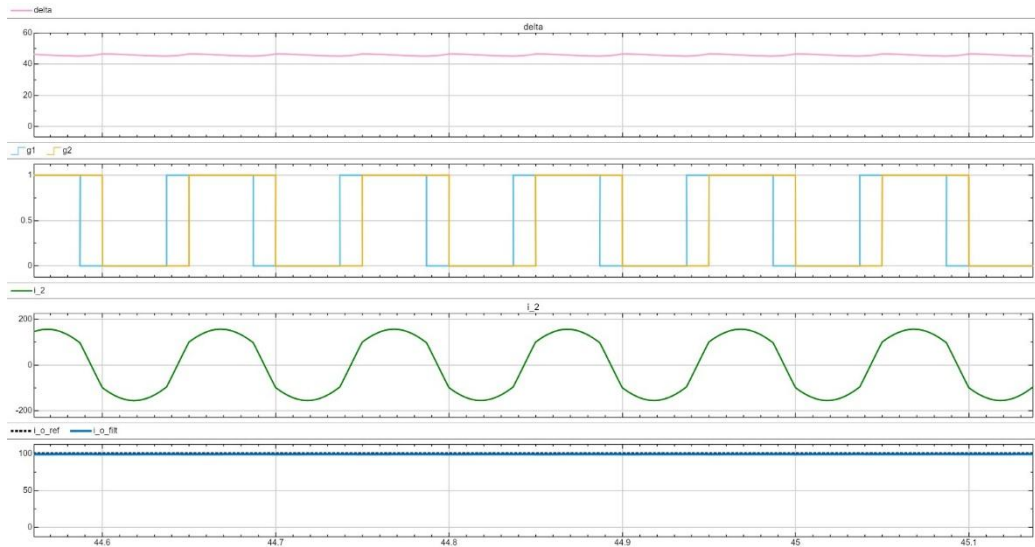


Figure 4: close-up of DAB waveforms

B. Single Active Bridge Converter

1) Theory of Operation

Single Active Bridge (SAB) is a simplified version of Dual Active Bridge (DAB). Here, we keep the full-bridge inverter on the primary side, but on the secondary side, we replace it with a diode bridge. We use a diode bridge to convert the AC signal from the transformer to DC. An LC filter is used to filter the output.

The transformer is also capable of providing isolation and converting the level of voltage, similar to that of the DAB. The same control method is used in the primary side, which is phase-shifted PWM with a duty cycle of 50%. The transferred power depends on the phase shift between the low-level switching signals.

The energy transferred between input and output can be calculated by the following equation:

$$P = \frac{V_1 V_2}{2\omega L} \sin(\delta)$$

Where:

- V_1 is the input voltage from the primary bridge
- V_2 is the output voltage after the diode rectifier and filter
- ω is the angular switching frequency
- L is the leakage inductance of the transformer
- δ is the phase shift angle

With each increase in phase shift angle δ , we transfer more power from the input to the output. The power travels in one direction only because the secondary side uses diodes instead of switches.

One of the advantages of SAB is lower cost, easier control, and fewer components. It does have several disadvantages, however. It cannot facilitate bidirectional power transfer, and it is harder to maintain soft switching (e.g., ZVS) at light loads.

2) Implementation

In order to achieve effective operation of the Single Active Bridge (SAB) converter in unidirectional control mode, an important adjustment was made on the secondary (output) side of the system. A series inductor was placed between the load and rectifier to filter the output current and provide smooth current tracking.

This extension was dictated by the natural character of rectifier circuits: conduction only occurs when the instantaneous AC voltage is greater than the DC output voltage. In the SAB topology, the transformer secondary acts as an AC source powering a passive full-bridge rectifier. For the current to be successfully conducted, the rectified waveform must be greater than the output voltage at its peak.

By keeping this guideline in mind, the system was optimized such that the secondary voltage swing is maintained sufficiently through phase-shift modulation and transformer design, so that the current conduction is started early and can last on during operation.

The value of the output inductor was selected from the target current ripple by the following equation:

The output inductor value was selected based on the target current ripple Δi , using the standard expression:

$$L = \frac{V_{out}(1 - D)}{f\Delta i}$$

Where:

f - switching frequency and D is the effective duty cycle or modulation ratio. This provided a low-ripple, high-resolution output current without the requirement for a second active bridge, so it was easier to design for the performance requirements. As shown in figure 5:

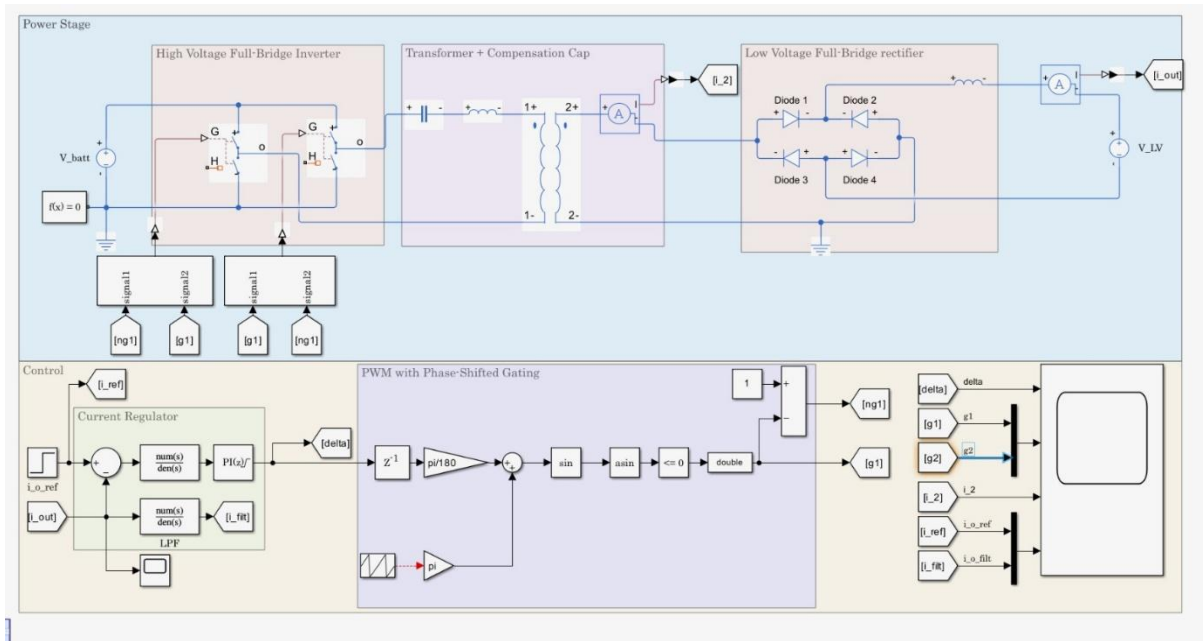


Figure 5: SAB implementation

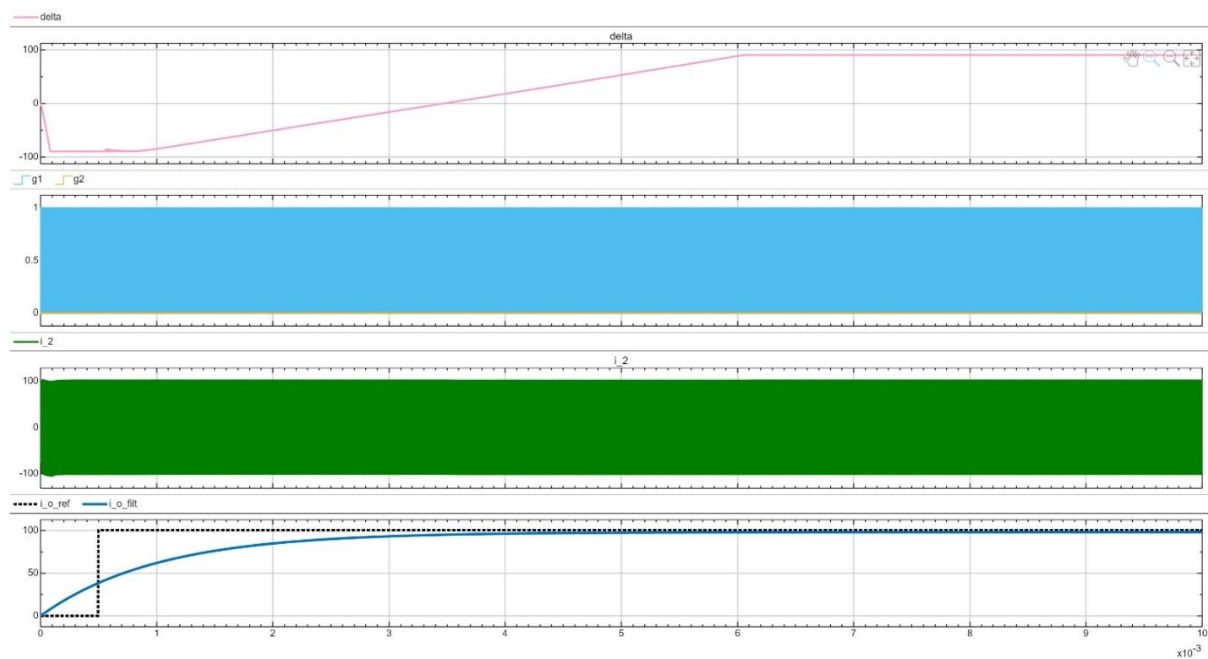


Figure 6: SAB waveforms

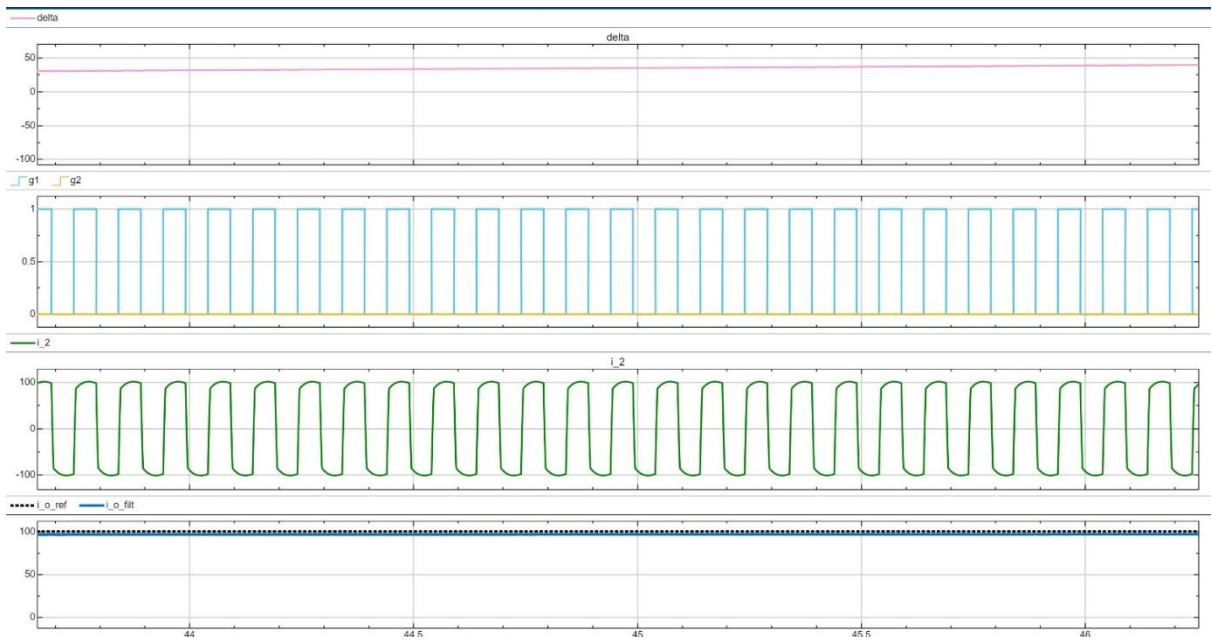


Figure 7: SAB waveforms close-up

III. CONCLUSION

Increased power conversion in auxiliary power modules (APMs) becomes increasingly important with requests from more powerful, more compact, and more power-conversion-efficient circuits being driven by new electric vehicle applications. The market, up to this point, has been characterized by showcase applications such as leveraging the utilization of Dual Active Bridge (DAB) and Single Active Bridge (SAB) converters, where higher frequency isolation and scalable and controllable power transfer are realized.

One of them, and worthy of note, is the DAB converter, which can send or receive power to or from the battery depending on driving or charging circumstances. This makes application to EV uses possible. The SAB converter, though being unidirectional, offers a direct solution to space- or direction-of-power-flow-limited applications.

Prime among them are also the efficient applications of the most popular APMs since they are easily compatible with soft switching techniques such as Zero Voltage Switching (ZVS). Having no hard switching, soft switching places critical components at zero voltage or current with very high accuracy, reducing loss, dissipating less heat, and protecting them from wear and tear. In EV APMs, space cannot be saved, and with regular cooling, it doesn't matter, and with efficiency being a very important factor within the vehicle's operating range and performance, this becomes particularly critical.

Therefore, soft-switching SAB and DAB converters are facilitating more powerful, lighter, and more complex electric vehicles as opposed to being a design breakthrough. The future transportation sector is going to be powered by both these technologies as EV design extends even further along the lines of further integration and higher efficiency.

IV. REFERENCES

- [1] J. V. G. Rama Rao and S. Venkateshwarlu, "Soft-switching dual active bridge converter-based bidirectional on-board charger for electric vehicles under vehicle-to-grid and grid-to-vehicle control optimization," *Journal of Engineering and Applied Science*, vol. 71, no. 49, pp. 1–16, 2024. [Online]. Available: <https://doi.org/10.1186/s44147-024-00384-z>.